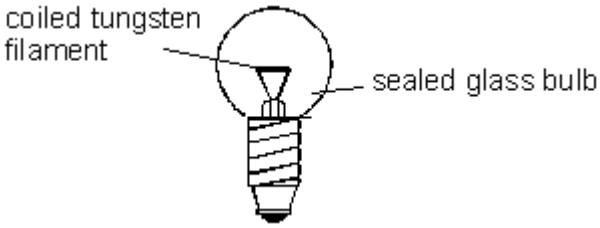
Q1.



The drawing shows a bulb from a torch. When the torch bulb is connected to a battery it lights up.

(a) What happens to the temperature of the filament when the torch is first switched on?

.....

1 mark

(b) Energy from the battery is transferred to the bulb. The bulb transfers the energy to the surroundings.

State **two** ways in which energy is transferred from the torch bulb to the surroundings.

1. .....

2. .....

2 marks

(c) Ravi accidentally breaks the glass of the bulb in his torch. He notices that the filament is not broken, so he switches the torch on.

The filament immediately 'burns out'.

(i) Which gas in the air is needed for the filament to burn?

.....

1 mark

(ii) Suggest how the makers of light bulbs prevent filaments from 'burning out' inside sealed bulbs.

.....

.....

1 mark

Maximum 5 marks

Q1.

(a) rises **or** increases

*accept 'goes up' **or** 'gets hot' **or** 'gets warmer'*

1 (L6)

(b) by thermal transfer

1 (L6)

by light

1 (L6)

answers may be in either order

*accept 'heat' **or** 'heating the air' **or** 'conduction' **or** 'convection' **or** 'by air molecules*

accept 'conduction and convection' for both marks

*accept 'radiation' **or** 'infra red'*

(c) (i) oxygen

accept 'O_2'

*do **not** accept 'O'*

1 (L6)

(ii) answers should give a reasonable way of providing an inert atmosphere in the bulb

any **one** from

- no oxygen inside bulb
- inert **or** noble **or** named noble gas inside bulb

*accept 'vacuum **or** no air inside bulb'*

*accept 'nitrogen **or** carbon dioxide inside bulb'*

*do **not** accept 'air tight'*

1 (L6)

[5]